Mission Statement

Mission: To organize and digitize vehicle violation and impoundment data, making it instantly accessible to police officers, government employees, and vehicle owners through a reliable web-based platform.

Vision: To become the standard digital infrastructure for traffic violation and vehicle impoundment management across Jordan.

Objectives & Goals

Goals

Eliminate paper-based violation recording in Jordanian traffic police operations.

Provide a reliable digital alternative when hardware violation devices fail due to extreme weather conditions such as high temperatures in desert and border areas.

Reduce the cost of maintaining and replacing dedicated hardware violation devices across police units.

Enhance transparency between traffic authorities and vehicle owners.

Objectives

1	**100% Paper Reduction** Within **3 months** of deployment via web-based alternative accessible when dedicated hardware violation devices fail due to battery drain or extreme weather conditions such as high temperatures in desert and border areas .
2	**Under 2 Minutes** Officers record violations digitally via browser — immediate backup when devices overheat or lose battery.
3	**60% Cost Decrease** Hardware maintenance and replacement costs reduced within 1 year of deployment, by reducing dependency on dedicated violation devices through the web-based system.
4	**Achieve system availability of 99% uptime within 6 months of deployment.**
5	**Allow vehicle owners to view their violations and payment status online within 24 hours of recording by first release date.**

SWOT Analysis

STRENGTHS

- Web-based — accessible from any device, no dedicated hardware needed
- Multi-role support: Admin, Police, Employee, Viewer
- Real-time violation recording and instant data storage
- Reduces dependency on paper and physical devices
- Vehicle images uploaded and stored digitally

WEAKNESSES

- Requires stable internet connection to function
- No mobile app — web only
- Basic authentication — no JWT or advanced security
- Local SQLite database — not cloud-based
- Limited to violation and impoundment management only

OPPORTUNITIES

- No similar unified system currently exists in Jordan
- Can be expanded to integrate with court and government systems
- Potential to cover all Jordanian governorates
- Could reduce operational costs across traffic police units
- Future mobile app development possible

THREATS

- Resistance from police officers to adopt new technology
- Hardware devices may still be preferred by some units
- Cybersecurity risks and unauthorized access
- Government budget limitations may delay deployment
- Internet outages in remote desert and border areas

Why Are We Doing This Project?

The Core Problem

Traffic police officers in Jordan currently depend on dedicated handheld hardware devices to record vehicle violations. These devices are highly vulnerable to failure caused by extreme environmental conditions — particularly the intense heat in desert regions and border areas such as the **Jordan-Saudi border, where temperatures regularly exceed 45°C** during summer months.

When devices fail due to battery drain or overheating, officers are forced to revert to **paper-based recording** — slow, error-prone, and difficult to manage at scale.

⚠ No unified digital system currently exists in Jordan for recording violations, managing impoundments, and enabling online owner access — without visiting a police station.

What This System Solves

Web-Based Backup

Accessible from any internet-connected device when hardware fails

Zero Paper

Eliminates paper dependency entirely across all police units

Cost Reduction

Reduces recurring hardware distribution and maintenance costs

Future Mobile App

Planned mobile application to further extend officer and owner accessibility

Project Scope

MUST DO

- Record traffic violations digitally via web browser
- Manage vehicle impoundment and lot assignment
- Multi-role access: Admin, Police Officer, Employee, Vehicle Owner
- Allow vehicle owners to view violations and payment status online
- Vehicle and owner registration and search

SHOULD DO

- Payment tracking and record closure
- Vehicle image upload and digital storage
- Police officer management and assignment

COULD DO

- Integration with Jordanian court systems
- Future mobile application for police officers
- SMS/email notification to vehicle owners

WON'T DO

- GPS real-time tracking of officers
- Online payment gateway
- Integration with external government databases

Operational Feasibility

Question	Answer
Does management support the project?	Yes — the system directly supports government traffic management goals
Do users support the project?	Police officers benefit from a reliable digital backup when hardware fails
Is the current system well liked?	No — paper-based fallback is widely considered unreliable and slow
Do users see the need for change?	Yes — hardware device failures in extreme heat create urgent demand
Will it require workforce reduction?	No — the system supports existing staff, not replaces them
Will training be required?	Minimal — the web interface is role-based and simple to use
Any legal or ethical concerns?	Data privacy of vehicle owners must be protected; access is role-restricted

Technical Feasibility

Hardware Requirements

Component	Specification
Application Server	Any machine running Java 17+ (min. 2GB RAM)
Client Devices	Any device with a modern web browser
Storage	Local disk for vehicle images (min. 50GB)
Network	Stable internet at police stations and checkpoints

Software & Technology Stack

Backend

Java Spring Boot 3.x with Apache Tomcat (embedded)

Database

SQLite (dev) → MySQL / PostgreSQL (production)

Frontend

HTML5 / CSS3 / JavaScript

API

RESTful API via Spring @RestController

File Handling

Spring Multipart File Upload

Build & IDE

Maven · NetBeans / IntelliJ IDEA

Economic Feasibility

Costs

Cost Item	Estimate
Development (3 team members)	Academic project — no direct labor cost
Hardware (server)	~$200–$500 one-time (basic server or cloud VM)
Software licenses	$0 — all open-source (Spring Boot, SQLite, Java)
Training	Minimal — simple UI requires < 1 hour orientation
Maintenance	Low — web-based updates require no device visits
Cost of NOT developing	Continued paper dependency, ongoing hardware failure costs, no owner transparency, potential legal disputes due to lost records

Benefits

Tangible

- Eliminate paper-based recording
- Reduce hardware device maintenance costs by **60%**
- Faster violation recording (< 2 minutes)
- Improved data accuracy and record keeping

Intangible

- Vehicle owners access records without visiting police station
- Enhanced government image and transparency